

Section 1: Multiple-choice questions (MCQs). [5 Marks]

Question 1: Mark the correct Answer (1/2 mark for each)

1. What is a combinational circuit?
 - a) A circuit whose output depends only on the current input
 - b) A circuit that has memory elements
 - c) A circuit that performs sequential operations
 - d) A circuit that uses feedback to produce outputs
2. A sum of product expressions is a product term (min term) or several min terms ORed together.
 - a) True
 - b) False
3. Which of the following is an incorrect SOP expression?
 - a) $x+x.y$
 - b) $(x+y)(x+z)$
 - c) x
 - d) $x+y$
4. The number of min terms for an expression comprising of 3 variables?
 - a) 8
 - b) 3
 - c) 0
 - d) 1
5. Which operation is shown in the following expression:
 $(X+Y').(X+Z).(Z'+Y')$
 - a) NOR
 - b) ExOR
 - c) SOP
 - d) POS
6. How many 4x16 Decoder using 2x4 Decoders
 - a) 4

8. In a 32-to-1 multiplexer, how many select lines are required?
I

a) 2

b) 3

c) 5

d) 6

9. What type of circuit is a half adder?

a) A sequential circuit

b) A combinational circuit

c) An analog circuit

d) A linear circuit
I

10. Decoder is a combinational circuit that accepts n inputs and produce $2 \times n$ outputs

a) True

b) False

c) I don't know

Section 2: Practice Design and Derive logic expressions

Question 2 [2 Marks]

Simplify by using De Morgan's Law $(A'B + BC')' \cdot (A \cdot B')'$

Question 3..... [6 Marks]

Implement the following Boolean function $F(A, B, C, D) = \sum(4, 5, 7, 9, 11, 12, 13, 15)$ using **4:1 mux**.

- a. Find F using truth table
- b. Design of the Multiplexer
- c. Implement the following function using the appropriate Decoder
 $F = \sum(0, 1, 4, 5)$

Question 4 [2 Marks]

Design the circuit from the following Boolean Expression by Using NOR Gates only

$$F = (A + B)(C + D)E$$

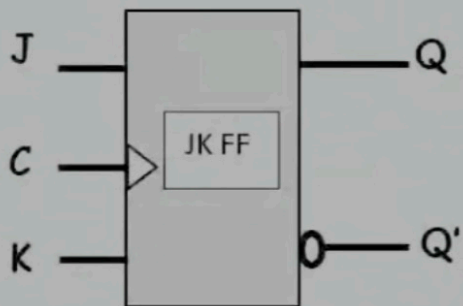
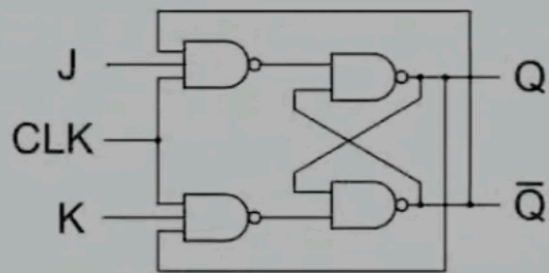
Question 5 [4 Marks]

Design a Full Adder using OR gate and 3x8 active high Decoder

Section 3: Sequential Circuit [6 Marks]

Given the following positive Edge Triggered JK Flip-flop.

- 1) Write the characteristic equation Q_{n+1} of the JK Flip-Flop then Complete the column of Q_{n+1} and state Description
- 2) Given the waveforms for CLK, J, and K, draw the corresponding outputs for Q and Q' of a JK Flip Flop



CLK	J	K	Q _n	Q _{n+1}	State Discription
0	X	X	0		
0	X	X	0		
↑	0	0	0		
↑	0	0	1		
↑	0	1	0		
↑	0	1	1		
↑	1	0	0		
↑	1	0	1		
↑	1	1	0		
↑	1	1	1		

Answer

1) Characteristic equation

2) Given the waveforms for CLK, J, and K, draw the corresponding outputs for Q and Q' of a JK Flip Flop